

Compressed certificates for unknot recognition

Arithmetic overlap families, regular-cover gluings,
and normal disks

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Abstract

We develop three exact computational improvements motivated by the remaining gap between the maintained **fastunknot** implementation and a general $n^{O(\log n)}$ recognition bound. First, a short endpoint word can identify an arithmetic progression containing every possible suffix–prefix overlap. An exact periodicity certificate makes candidate validity monotone in progression rank; galloping and bisection filter the entire progression. Its step and cardinality may both be exponentially large in their binary descriptions. This strictly extends the implementation’s fixed short-period and bounded-marker shortcuts while preserving the complete fallback.

Second, we give a polynomial algorithm, in the encoded input size, for equivalence of framed assemblies of connected regular cyclic or dihedral surface covers. A spanning forest reduces all local gluing phases to one group element per graph component. Cycle and boundary-mark constraints become at most two systems of congruences. We also count the remaining unmarked classes exactly: cyclic assemblies have m^β classes, where β is the cycle rank. Thus efficient equivalence testing alone does not control the number of distinct geometric states.

Third, we implement an independent normal-disk verifier using a primitive extreme-ray certificate, modular rank, Euler characteristic, and boundary intersection parity. It validates the finite triangulation and never expands the normal surface. Fibonacci layered solid tori provide an explicit exponential-weight test family. These results include proofs, adversarial checks, measured positive and negative outcomes, and a repository patch. The general quasi-polynomial recognition bound remains unproved.

Research prepared for integration into <https://github.com/VladimirReshetnikov/ProveIt>.

All new numerical claims are tied to reproducible artifacts. Classical normal-surface and compressed-word results are credited separately from the implementation and refinements developed here.

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1 Research status and the concrete advance

1.1 What is being continued

The maintained repository already contains a substantial exact recognizer: structural and braid certificates, polynomial one-sided obstructions, width-sensitive Jones and Khovanov computations, compressed group operations, and an optional complete normal-surface backend. Its synthesis article had reached 210 pages before this work. Repeating the existing width-conditional recurrences, or adding another invariant with no effect on the surviving search space, would not close its main theoretical gap.

The public checkout and preceding relevant revision are pinned as follows [1]:

Checkout	1be2abc8cc743c1a3b5cb7fdfeb650697f8e2d5b
Subtree revision	38e65c9c2e2bfd42b9f71f1a1a7af15a3f8a3005
Shared <code>fast/</code> tree	e722244bb0d7f35cf7f34f83228b350240e4d13

The preceding subtree revision is entitled “Document sparse endpoint proofs and controlled overlap measurements.” Both references have the same maintained `fast/` tree. The patch is prepared against the full checkout reference. It does not depend on remembering an earlier conversational archive.

The latest compressed-overlap implementation already had two certified shortcuts. One recognized a common cyclic pattern with period at most 32. The other counted occurrences of an endpoint letter, saturating at 65, and individually verified at most 64 forced candidates. Their exact fallbacks were complete, but long-period blocks with many repeated markers could still exhaust the allotted work. We remove those two numerical restrictions on a structurally certified class. The important change is from enumerating a bounded number of candidate lengths to proving a formula for all of them.

Two geometric interfaces are developed alongside that change. The existing surface-cover kernel describes a particular affine action on a cyclic sheet set. It does not solve gluing equivalence for an assembly of regular covers. The new assembly kernel has a deliberately different, checked input model. The existing Regina wrapper exports an external-engine verdict and metadata; it does not independently certify a supplied normal disk. The new verifier checks such a disk in a supplied finite triangulation. These are exact additional capabilities, with their premises visible at the interface.

1.2 What the current literature actually supplies

Lackenby’s July 2026 preprint gives a new algorithm based on essential hierarchies and boundary patterns [2]. Section 9 explicitly distinguishes a bound on the number of steps from a running-time estimate for all steps. Its later sections develop polynomial verification and compressed decomposition when suitable surfaces are supplied. In particular, supplying a normal surface in binary coordinates and cutting efficiently along it are not equivalent to finding the required surfaces or bounding the complete recognition computation. The repository correctly preserves this distinction.

Compressed-word equality and several fully compressed matching operations already have polynomial algorithms in grammar size [5, 6, 7]. The present endpoint construction is a more economical exact route for a broad promised class of overlap queries. We do not claim that it changes a previously exponential general compressed matching problem into a polynomial one.

Likewise, polynomial normal-surface verification is classical. The extreme-ray approach is part of the Hass–Lagarias–Pippenger framework [3]; the more general orbit-counting machinery of Agol–

Hass–Thurston handles surfaces beyond this restricted certificate class [4]. Our contribution is a small independently inspectable implementation, an explicit modular rank certificate, a parity test for the boundary, and a reproducible comparison that does not require expanding a normal disk. General normal-surface discovery remains a separate problem.

1.3 Results and their precise scope

Result	Exact guarantee	Remaining premise or limitation
Endpoint progressions	All suffix–prefix overlaps are returned from a certified short-anchor occurrence progression; both its step and count are binary integers with no fixed cap.	Failure of the structural certificate uses the complete matcher; the inherited extension queries can still be expensive.
Regular-cover gluing	All framed isomorphisms are encoded by at most two root congruence classes per connected assembly graph.	Each block is a connected regular cover with checked compatible whole-boundary attachments; the blocks are already supplied.
Gluing state count	Exact orbit counts, including m^β for cyclic covers, quantify states left after relabelling.	This is a lower bound for enumerating these framed classes, not for arbitrary unknot algorithms.
Normal disk certificate	A supplied finite triangulation and binary normal vector are checked to contain an essential compressing disk.	The vector must satisfy the extreme-ray condition; correspondence with a given PD diagram is not asserted.

The distinction between a complete recognizer and a complete local operation is central. The word shortcut participates in an already existing optional compressed group search. The gluing and normal-disk kernels are new explicit APIs. They do not silently introduce new positive or negative knot verdicts into the pipeline.

2 Encodings, cost measures, and proof obligations

The number n denotes crossings in a validated classical knot diagram, not expanded word length or the size of an unencoded geometric construction. We use the specific target $n^{O(\log n)} = 2^{O((\log n)^2)}$. Some authors use “quasi-polynomial” for the larger union of bounds $2^{(\log n)^{O(1)}}$; any use of that broader convention must be stated.

An SLP is an acyclic binary grammar whose rules are terminals or concatenations of earlier rules. Let g be the number of reachable rules, H its height, L the maximum represented length, and $B = \lceil \log_2(L + 1) \rceil$. Lengths, positions, and counts are binary integers. Signed generator names have their own bit lengths and are compared exactly. A grammar may contain two copies of the same child in one rule; its represented occurrence count must therefore respect multiplicity even when the grammar nodes themselves are shared.

For surface assemblies, m is the cyclic parameter, and $\ell = \lceil \log_2(m + 1) \rceil$. The regular dihedral group has order $2m$; it is not the generally nonregular action of a dihedral subgroup on m residues. Let v, e, r denote the numbers of blocks, attachments, and explicit markings, with the

total listed local monodromy data included in the input size. For a connected assembly graph, $\beta = e - v + 1$.

For the normal-disk verifier, t is the number of tetrahedra, x the normal coordinate vector, $s = |\text{supp } x| \leq 7t$, and B_x the largest coordinate bit length. A tetrahedron has four triangle and three quadrilateral types. Face maps are explicit vertex permutations, not an unverified topology label or a hash of one.

We distinguish three measurements throughout. A mathematical bit bound counts integer operations according to operand length. The implementation's work counter charges selected operations cooperatively; it is a resource policy, not an exact instruction count. Wall-clock measurements are empirical and include the scope specified by each benchmark. A resource-limited computation has no recorded completion time. It cannot serve as the denominator of a completion-speedup claim.

Proposition 2.1 (Local bounds do not compose without state bounds). *Suppose each operation on an encoded state of size S costs S^c for a fixed c . If a computation performs $R(n)$ charged state-processing operations, including repeated visits, and every participating state has size at most $S(n)$, its operation cost is bounded by $R(n)S(n)^c$, before adding input conversion and output verification. To obtain the target $n^{O(\log n)}$ from this estimate, it suffices to prove both $\log R(n) = O((\log n)^2)$ and $\log S(n) = O((\log n)^2)$, with the other work similarly controlled.*

Proof. Sum the local cost over the actual operations. Taking logarithms yields $\log R + c \log S$. The statement requires bounds for the states actually processed and the number of processing operations; the existence of a smaller representation, or a bound only on distinct states, is insufficient. $\square$

This elementary accounting is included to locate the obligation, not as a new recognition theorem. The gluing orbit count below gives a concrete reason that an efficient quotient computation can leave too many distinct states for this argument to meet the target.

3 All overlaps from an arithmetic endpoint family

3.1 Candidate lengths forced by a short endpoint word

For finite words X, Y , define

$$\mathcal{O}(X, Y) = \{k : 1 \leq k \leq \min(|X|, |Y|), \text{suff}_k(X) = \text{pref}_k(Y)\}.$$

These are literal signed-word equalities. Free reduction is not performed inside this definition. In the application, the surrounding group algorithm chooses the words whose overlaps are to be queried.

Write $L = \min(|X|, |Y|)$. Replacing X by its last L letters and Y by its first L letters preserves $\mathcal{O}(X, Y)$. We therefore assume both operands have length L . Pick $1 \leq q \leq L$ and let $P = \text{suff}_q(X)$. Every overlap of length $k \geq q$ forces an occurrence of P in Y ending at position k , where positions are counted from one. Thus the set

$$K_P(Y) = \{i + q : Y[i : i + q] = P\}$$

contains every possible overlap length at least q . This is only a necessary condition. The algorithm must verify all lengths that it accepts.

An arithmetic progression is represented by (a, d, c) , denoting $\{a, a + d, \dots, a + (c - 1)d\}$. For $c > 1$, require $d > 0$; for a singleton, use $d = 0$. Both d and c are exact binary integers.

Theorem 3.1 (Endpoint progression clipping). *Suppose $K_P(Y)$ is a nonempty arithmetic progression K with largest member M , and put $U = \text{pref}_M(Y)$. If K has at least two members, let its step be d and suppose that U has period d . Then*

$$\mathcal{O}(X, Y) \cap [q, L] = K \cap [q, \text{lcsuf}(X, U)]. \quad (1)$$

The hypothesis about the period is decidable by the exact equality

$$U[0 : M - d] = U[d : M]. \quad (2)$$

No periodicity hypothesis on the other operand X is required.

Proof. If $K = \{M\}$, the only candidate is accepted exactly when $\text{suff}_M(X) = U$, equivalently when $M \leq \text{lcsuf}(X, U)$. Now suppose K has at least two members. Each $k \in K$ has the form $M - jd$ for an integer $j \geq 0$. The period condition implies $U[i] = U[i + jd]$ whenever both sides are within U , by repeatedly applying the period identity. Therefore

$$\text{pref}_k(Y) = \text{pref}_k(U) = \text{suff}_k(U).$$

The condition $\text{suff}_k(X) = \text{pref}_k(Y)$ is consequently equivalent to $\text{suff}_k(X) = \text{suff}_k(U)$, which holds precisely when $k \leq \text{lcsuf}(X, U)$. Conversely, every overlap of length at least q belongs to K by its final q letters. These two implications prove (1). Equation (2) is the definition of period d for a finite word, so it gives an exact certificate for the only periodicity used in the proof. $\square$

Corollary 3.2 (Symmetric endpoint direction). *Let $Q = \text{pref}_q(Y)$ and let $K = \{L - i : X[i : i + q] = Q\}$. Suppose K is a nonsingleton arithmetic progression of step d and maximum M , and $V = \text{suff}_M(X)$ has period d . Then*

$$\mathcal{O}(X, Y) \cap [q, L] = K \cap [q, \text{lcp}(V, Y)].$$

Proof. For $k = M - jd$, periodicity gives $\text{suff}_k(X) = \text{suff}_k(V) = \text{pref}_k(V)$. Now use the definition of a longest common prefix. The occurrence of Q at the beginning of any overlap proves necessity of membership in K . $\square$

Empty and singleton candidate sets need separate, simpler treatment. If the set is empty, no overlap of length at least q exists. For a singleton, one exact suffix–prefix comparison decides its membership. Every length below q is checked directly in all three cases. Hence one successful anchor returns at most q disjoint progression records: at most $q - 1$ short singletons and one progression for the longer overlaps.

3.2 Finding the cutoff by candidate rank

The common-suffix and common-prefix lengths in the preceding statements describe the answer, but need not be computed. A binary search for the complete common-prefix length can examine positions that are irrelevant to the progression. Once the progression and its period have been checked, we can search only its candidate ranks.

Corollary 3.3 (Galloping search for the last successful candidate). *Under the hypotheses of theorem 3.1 or corollary 3.2, write $K = \{a + jd : 0 \leq j < c\}$ with $c \geq 2$. Let h be the number of its members that are actual overlaps and let $z = c - h$. All accepted members can be returned using*

$$O(1 + \log(1 + z)) \quad \text{exact suffix–prefix equality comparisons.} \quad (3)$$

This is $O(\log(c + 1))$ in the worst case. If every candidate succeeds, one comparison suffices; if every candidate fails, at most two suffice. These counts exclude the preceding period check and the fewer than q short-length checks.

Proof. By the clipping theorem, the predicate

$$f(j) = [\text{suff}_{a+jd}(X) = \text{pref}_{a+jd}(Y)]$$

is a true prefix: it holds precisely for $0 \leq j < h$. Check $f(c-1)$ first. If it holds, return all of K . Otherwise check $f(0)$; if that fails, return the empty progression. The remaining case has $1 \leq h < c$. Maintain a checked true index lo and a checked false index hi, initially 0 and $c-1$. Probe backward from the maximum at distances $\delta = 1, 2, 4, \dots$, using $j = \max(0, c-1-\delta)$. A false answer updates $\text{hi} = j$; a true answer updates $\text{lo} = j$ and ends this phase. If $j = 0$, its previously checked true value supplies the endpoint without another comparison. For an unclamped probe,

$$f(c-1-\delta) \text{ holds} \iff \delta \geq z.$$

Consequently the gallop uses $O(1 + \log z)$ comparisons. Its final bracket has width $O(z)$, including the clamped case. Binary search within that bracket finds the last true index using $O(\log(1+z))$ additional comparisons. The answer is the first h members of K , represented by one progression, or by a singleton when $h = 1$. The invariant also covers $c = 2$, when no interior search is needed. $\square$

The monotonicity is a consequence of the certified period condition. Arbitrary overlap lengths need not have this property. The implementation therefore applies the rank search only after the exact structural check; it does not replace a failed period test by an assumed monotonicity. The proof uses ordinary exponential search followed by bisection, rather than a new general-purpose search method. Its useful feature here is the choice of the candidate rank and the direction of the gallop.

Example 3.4 (The opposite word may be nonperiodic). Let B be a word whose final letter c appears nowhere else in B . For integers $C \geq r \geq 1$, let $Y = B^C$ and let $X = WB^r$, where W is any word of length $(C-r)|B|$. The endpoint occurrences in Y lie at $|B|, 2|B|, \dots, C|B|$. The theorem returns exactly those multiples no larger than the actual common-suffix length. It does not need to certify that W repeats B , and it remains correct when W contains arbitrary defects or a partially matching suffix.

3.3 Recognizing a progression without listing its points

The missing operation is to summarize all occurrences of an explicit short pattern in an SLP word. It is insufficient to summarize distinct grammar nodes containing the pattern: a shared node can be used exponentially many times in the represented word.

For an ordered finite set of occurrence starts S , retain

$$\Sigma(S) = (c, f, l, \delta_{\min}, \delta_{\max}),$$

where $c = |S|$, f, l are its endpoints, and the last two entries are the minimum and maximum consecutive gaps. Empty and singleton sets have both gap entries zero. For $c \geq 2$,

$$S \text{ is an arithmetic progression} \iff \delta_{\min} = \delta_{\max}. \quad (4)$$

Lemma 3.5 (Exact ordered concatenation). *Suppose every point of S precedes every point of T . The summary of $S \cup T$ is determined by their summaries: counts add, the first and last nonempty endpoints persist, and the new gap extrema are the extrema among the old gaps and $\min T - \max S$. Translating T only changes its two endpoints. These formulas remain exact when the sets are exponentially large.*

Proof. All consecutive pairs in $S \cup T$ lie within S , within T , or are the single pair bridging their endpoints. This exhausts the gap list. $\square$

For a rule $A \rightarrow BC$ and pattern length q , every occurrence is in exactly one of three disjoint sets: wholly inside B ; crossing the concatenation cut; or wholly inside C . The last set is translated by $|B|$. This locality at a grammar concatenation is standard in compressed q -gram algorithms; see Goto, Bannai, Inenaga and Takeda [9]. The five-entry summary used here is tailored to certifying one exact occurrence progression. There are at most $q - 1$ crossing positions. Store the first and last $\min(q - 1, |A|)$ letters of every reachable rule, and check those crossing positions explicitly. Processing them in increasing start order allows [lemma 3.5](#) to merge all three sets without enumerating any child occurrence set. For $q = 1$ there are no crossing occurrences.

Theorem 3.6 (SLP occurrence summaries). *For an explicit nonempty pattern of length q and an SLP with g reachable rules, the occurrence summary can be computed using $O(qg)$ stored letters and integers and $O(q^2g + g \log(g + 1))$ elementary operations with the implementation's topological sorting. If b bounds the bit length of positions, counts, rule identifiers and signed letters, a conservative corresponding bit bound is $O((q^2g + g \log(g + 1))b)$, besides the input grammar's stored structure.*

Proof. Terminal rules are immediate. The three-way partition at a concatenation and [lemma 3.5](#) establish correctness by induction on the grammar. The retained endpoints have length at most $q - 1$. There are at most $q - 1$ crossing candidates, each checked against q explicit letters, yielding $O(q^2)$ elementary work per rule. The count for a repeated child is included once on each side of the cut, correctly accounting for grammar sharing. All integer additions, subtractions, comparisons and translations involve $O(b)$ bits. Sorting the reachable rule identifiers adds the displayed term; node order already respects the grammar's acyclicity. $\square$

The summary is useful even when [equation \(4\)](#) fails: it certifies that this particular anchor does not give a single progression. It is not a complete compressed representation of a nonprogression set. The matcher therefore tries another anchor or falls back, rather than discarding unrepresented candidates.

3.4 Integration and local complexity

The implementation tries anchor sizes 1, 2, 4, 8, in both directions. Its public primitive accepts other finite lists of positive sizes. A proposed size is clipped to the available word length. The integrated hook is reached only for a query window of length at least 128, after the earlier uniform, short-period and sparse-endpoint shortcuts. Thus successful earlier paths retain their previous work. The new hook is deterministic and preserves the old complete dyadic occurrence-table fallback.

For one nonsingleton successful anchor, the implementation performs one period equality and the candidate-rank comparisons in [corollary 3.3](#). It does not compute the complete longest common prefix or suffix. Let g count the rules reachable from the input operands, let H bound their height, and let b bound the bit lengths of all positions, counts, rule identifiers and signed letters involved in the query. Write $E(G, b)$ for a bound on one exact equality call on an augmented grammar with at most G rules, including its assertion and cache bookkeeping. To account also for persistent arena data, let g_0 bound the total number of grammar rules and pre-existing equality-cache records at entry; in particular $g_0 \geq g$. Define

$$J = 1 + \lceil \log_2(1 + z) \rceil.$$

Each candidate comparison slices the same original operands. A substring adds $O(H)$ rules and does not increase their height, so the initial cropping, endpoint extraction, period slices, candidate slices and short checks together create at most $O(H(q + J))$ additional rules. The equality implementation itself uses assertions on existing rules. We may therefore take

$$G = g_0 + O(H(q + J)).$$

This deliberately counts persistent data that a reachable-rule summary does not inspect but an equality-cache cleanup might visit. Charging $O(b^2)$ for elementary binary multiplication in the calculation $a + jd$, a conservative bound for the new promised-class route is

$$O\left((q^2g + g \log(g + 1))b + (q + J)[E(G, b) + Hb + b^2]\right). \quad (5)$$

The q term includes the fewer than q directly checked short lengths. For the fixed list 1, 2, 4, 8, failed earlier anchors contribute only a constant number of summary passes, endpoint slices and period comparisons; the same bound applies with the largest tried q and a changed absolute constant. All large numerical dependence is through encoded lengths and the exact comparison cost. In particular, [equation \(3\)](#) improves the *number* of comparisons; it does not make each equality independent of the represented words. No asymptotic bound is transferred from a different compressed-string implementation.

All calls use the same arena work allowance and cancellation callback. Successful complete answers are cached under the caller's original root pair, even when the actual comparison was cropped. Candidate-rank filtering uses direct suffix-prefix comparisons in both anchor directions. An interruption raises the existing resource exception, with no partial overlap set entered into the answer cache.

An unsuccessful speculative certificate has a bounded number of summary passes for the fixed policy, but its exact equality query may still be expensive. This is why the experiments include failed period tests and phase-shifted long-period queries. We claim the promised-class improvement expressed in [equation \(5\)](#), not uniform dominance over every old query.

3.5 An explicit family beyond both old numerical caps

Proposition 3.7 (Unbounded period and marker multiplicity). *For distinct signed letters a, b, c , positive integers N, C , and $W = ((ab)^N c)^C$, let $d = 2N + 1$. Then*

$$\mathcal{O}(W, W) = \{d, 2d, \dots, Cd\}.$$

The $q = 1$ endpoint method returns this set as a single progression, using grammar operations polynomial in $\log(N + 1) + \log(C + 1)$ for its constructed SLP. Neither d nor C has a fixed numerical upper bound.

Proof. The final letter is c , whose occurrences are exactly at multiples of d . Every overlap must end at such a position in the second copy. Conversely, each such length is a whole number of identical blocks and is an overlap. The longest candidate window is all of W and has period d . Repeated squaring gives a grammar with $O(\log(N + 1) + \log(C + 1))$ rules, and the exact compressed comparisons are polynomial in that encoded grammar size. $\square$

Two binary-alphabet variants demonstrate why short words are useful anchors:

$$W_2 = ((ba)^N a)^C, \quad W_4 = ((ab)^N aa)^C.$$

The terminal word aa identifies exactly the block ends of W_2 . The terminal word $abaa$ does so for W_4 . For $N \geq 2$ the resulting overlap sets are, respectively, the multiples of $2N + 1$, and

$\{1\}$ together with the multiples of $2N + 2$. The latter isolated overlap is exactly why lengths shorter than the anchor must be retained.

With $N = C = 2^{500}$, a period itself needs about 502 bits and the number of overlaps needs 501 bits. Explicitly listing the overlaps is impossible within reasonable resources, but their three-integer progression representation is small. This family is a controlled compressed-string benchmark. No theorem here says that arbitrary hard unknot instances must generate this family, or that every important relator overlap satisfies its hypotheses.

4 Regular-cover gluings as congruence problems

4.1 The geometric model and why regularity matters

Fix $G = C_m$ or $G = D_m$. We write dihedral elements as $(\epsilon, t) = T^t R^\epsilon$, where $\epsilon \in \{0, 1\}$, $t \in \mathbb{Z}/m\mathbb{Z}$, and

$$(\epsilon, t)(\eta, u) = (\epsilon \oplus \eta, t + (-1)^\epsilon u). \quad (6)$$

For the cyclic group only parity zero is allowed. The order of D_m is $2m$, including the small degenerate parameters $m = 1, 2$.

Each block is a connected regular cover of a specified compact connected orientable surface with nonempty boundary. Its fibre over a chosen basepoint is identified with the set G , and monodromy acts by left multiplication. The based peripheral system includes fixed paths from that basepoint to the boundary circles. These paths are part of the coordinate framing. The images of the supplied canonical free generators must generate all of G ; the implementation checks this by gcds and the presence of a reflection in the dihedral case. This condition makes the block connected and its monodromy action regular.

Any automorphism of one such block covering the identity on its fixed base surface acts on the fibre by right multiplication. Indeed, a fibre map f commuting with every left multiplication satisfies $f(g) = gf(1)$ for all $g \in G$. Write it $g \mapsto gh$. This observation is the reason one group element per block suffices. It does not hold in this form for an arbitrary nonregular cover.

An oriented attachment edge $e : u \rightarrow v$ joins distinct available boundary ports, possibly of the same block. Every port is used at most once. Choose compatible reference gluings reversing the base boundary orientations. In the fixed fibre coordinates, the two peripheral monodromies must be exact inverses; this is checked. A phase $a_e \in G$ then glues the entire preimage of one boundary circle to the entire preimage of the other by $g \mapsto ga_e$ at the reference fibre. This is well-defined because right multiplication commutes with the left peripheral action.

Thus attachments are allowed to permute the lifted boundary circles in a specified algebraic fashion. They are not arbitrary independently chosen permutations of individual lifts. The assembly graph is labelled and fixed, as are the local monodromies, base surfaces, and boundary ports. In particular, the natural dihedral action on m points is generally not the regular action on $2m$ points used here; substituting it would invalidate the one-group-element description of all local transports.

Remark 4.1 (Local and global regularity). Every block cover is regular. The cover of the assembled base surface need not be regular: compatible deck maps are constrained by the holonomy around assembly cycles. The word “regular” in this section always describes the local blocks and their available right multiplications.

4.2 Eliminating all tree phases

Consider two phase assignments a_e and a'_e on identical local block data. An isomorphism over the fixed labelled base decomposition consists of right multiplications h_v such that

$$a_e h_v = h_u a'_e \quad (e : u \rightarrow v). \quad (7)$$

This equation follows by applying the map before or after the attachment to an arbitrary fibre element. It is therefore both necessary and sufficient for the local maps to glue to a global covering isomorphism.

Choose a spanning forest. For a component root o , let A_v be the ordered product of original phases on the tree path from o to v , using inverses when an edge is traversed backwards. Define A'_v similarly.

Theorem 4.2 (Forest reduction of gluing equivalence). *Every solution of the tree-edge equations is determined uniquely by one root element $h = h_o$ and has the form*

$$h_v = A_v^{-1} h A'_v. \quad (8)$$

For a remaining edge $e : u \rightarrow v$, define

$$c_e = A_u a_e A_v^{-1}, \quad c'_e = A'_u a'_e (A'_v)^{-1}.$$

The remaining conditions are exactly

$$c_e h = h c'_e. \quad (9)$$

Proof. For a tree edge traversed from u to v , solve (7) for h_v and proceed along the unique tree path. Reversing an edge gives the inverse formula, so induction establishes (8) and uniqueness. Substitute (8) into (7), multiply on the left by A_u and on the right by $(A'_v)^{-1}$, and obtain (9). The same manipulations in reverse prove sufficiency on all edges. $\square$

All products here involve fixed-size group records and integers reduced modulo m . The algorithm never unfolds a fibre of m or $2m$ sheets. There is also no need to try $|G|^v$ possible collections of local maps.

4.3 At most two scalar congruence systems

For D_m , write $h = (\epsilon, t)$ and fix ϵ temporarily. If $c = (\eta, b)$ and $c' = (\eta', b')$, equation (9) first requires $\eta = \eta'$. If $\eta = 0$, it requires

$$b = (-1)^\epsilon b' \pmod{m}, \quad (10)$$

with no constraint on t . If $\eta = 1$, it requires

$$2t = b - (-1)^\epsilon b' \pmod{m}. \quad (11)$$

Let $d_2 = \gcd(2, m)$. The latter is inconsistent unless its right side is divisible by d_2 . Otherwise it is a single congruence for t modulo m/d_2 , obtained by dividing by d_2 and inverting $2/d_2$. Only the two choices of ϵ are tried. The cyclic case has only $\epsilon = 0$ and all cycle equations are equality tests of rotation shifts.

Markings can be included without adding a branching factor per mark. An ordered fibre-point mark $x \mapsto y$ at block v requires $x h_v = y$, and hence

$$h = A_v x^{-1} y (A'_v)^{-1}. \quad (12)$$

A lifted boundary circle is an orbit Hx of the peripheral subgroup $H = \langle p \rangle$. Requiring this named circle to map to Hy is equivalent to

$$xA_v^{-1}hA'_vy^{-1} \in H. \quad (13)$$

This condition preserves the specified circle without choosing a particular point on it.

If $p = (0, r)$ is a rotation, then $H = \{(0, jd) : j \in \mathbb{Z}\}$, where $d = \gcd(m, r)$. For fixed root parity, (13) first tests a parity and then is one congruence for t modulo d . If $p = (1, r)$ is a reflection, then $H = \{(0, 0), (1, r)\}$. The parity of the left side selects which of those two elements is possible, leaving one congruence modulo m . These derivations also handle the identity peripheral element: each boundary circle then contains just one point of the reference fibre.

Lemma 4.3 (Congruence intersection with an explicit obstruction). *The two conditions $t \equiv r \pmod{d}$ and $t \equiv s \pmod{e}$ are compatible if and only if $\gcd(d, e) \mid (s - r)$. If compatible, their intersection is one residue class modulo $\text{lcm}(d, e)$, obtainable by the extended Euclidean algorithm. If incompatible, the failed divisibility is a certificate of inconsistency.*

Proof. Substitute $t = r + dz$. The equation $dz \equiv s - r \pmod{e}$ is solvable exactly when the stated gcd divides its right side. Divide by the gcd to obtain an invertible coefficient. The solutions differ by $e/\gcd(d, e)$ in z , so by the lcm in t . $\square$

Theorem 4.4 (Complete marked gluing solver). *For two checked assemblies in the model above and any explicit collection of fibre-point and boundary-lift marks, all covering isomorphisms over the fixed labelled base decomposition can be encoded by at most two records per connected assembly graph. Each record has the form*

$$h_o = (\epsilon, r + jD), \quad 0 \leq j < m/D, \quad (14)$$

where $D \mid m$. Every local transport follows from (8). Empty branches have independently checkable contradiction records. The algorithm has polynomial bit complexity in the full listed input and $\log(m + 1)$.

Proof. The forest theorem is exact. For each possible root parity, (10)–(13) reduce every remaining requirement to a failed parity/equality test or a single congruence modulo a divisor of m . Repeated use of lemma 4.3 yields either a contradiction or precisely one residue class modulo a divisor D of m . Its representatives in $\mathbb{Z}/m\mathbb{Z}$ are (14). There are no other solutions, since every original equation or marking was included. Conversely, substitution proves that every reported family member gives an isomorphism. A forest pass and a linear number of gcd, modular inverse, and group operations use integers of $O(\log(m + 1))$ bits, together with the explicit graph and monodromy indices. Polynomial bit complexity follows. Components are retained as a product of solution families rather than enumerated jointly. $\square$

The implementation provides two replay routes. One checks a single collection of local transports directly against (7) and the original marks, without using a spanning forest or CRT. The other checks the reported complete solution families and their contradiction records. It verifies the congruences and their combined modulus, rather than rerunning the producer's CRT search. These routes separate the correctness of a particular witness from the stronger assertion that no isomorphisms were omitted.

For a more explicit bound, let N count the listed vertices, edges, free generators and marks, and let B include the largest input-integer bit length and $\log(N + 2)$. After input normalization, every root residue and modulus has $O(\log(m + 1))$ bits. A conservative schoolbook bound for a

connected graph is $O(NB^3)$ bit operations, with $O(NB)$ bits of representation. Constraint work is grouped by graph component, so the solution-family computation has the same bound for a disconnected graph. The implementation additionally reports the product of all component isomorphism counts; explicitly multiplying those binary counts costs at most another $O(N^2B^2)$ operations. No map family is enumerated.

4.4 Canonicalization and exact residual state counts

For an unmarked connected assembly, forest normalization makes the tree phases identities. The remaining β phases form a tuple in G^β ; the residual root change simultaneously conjugates this tuple. This gives both a canonical cache representation and a count of inequivalent framed assignments.

For C_m , conjugation is trivial. For D_m , if all tuple entries are rotations, compare the tuple of shifts with its simultaneous negation and choose the lexicographically smaller one. If there is a reflection, choose the first. Its shift a can be normalized to $\delta = 0$ when m is odd, and to $\delta = a \bmod 2$ when m is even. For root parity ϵ , solve

$$2t = a - (-1)^\epsilon \delta \pmod{m}.$$

There are at most two parity candidates for the resulting normalized tuple. When m is even, the two possible solutions differing by $m/2$ have the same conjugation action because the half-turn is central. Taking the smaller tuple therefore gives a polynomial-time canonical representative for this unmarked, fixed-base equivalence relation.

Theorem 4.5 (Orbit counts after all block deck changes). *On a fixed connected assembly graph of cycle rank β , with compatible local blocks and no additional marks, the number of phase assignments up to block deck changes is*

$$N_{C_m}(\beta) = m^\beta. \quad (15)$$

For the regular dihedral model it is

$$N_{D_m}(\beta) = \begin{cases} \frac{(2m)^\beta + (m-1)m^\beta + m2^\beta}{2m}, & m \text{ odd}, \\ \frac{2(2m)^\beta + (m-2)m^\beta + m4^\beta}{2m}, & m \text{ even}. \end{cases} \quad (16)$$

Proof. Forest normalization identifies the desired classes with simultaneous conjugacy orbits on G^β . This proves (15). For a finite group, double-count pairs consisting of a tuple and a group element fixing it: the number of orbits is $|G|^{-1} \sum_{g \in G} |C_G(g)|^\beta$. For odd m , the identity has centralizer size $2m$, the other $m-1$ rotations have centralizer size m , and each of the m reflections has centralizer size 2. For even m , the identity and half-turn have centralizer size $2m$, the other $m-2$ rotations have size m , and reflections have size 4. Substitution proves (16), including $m = 1, 2$ and $\beta = 0$. $\square$

Corollary 4.6 (When the assembled cover is regular). *For a connected assembly graph, without imposing additional marks, the deck group of the cover of the assembled base is isomorphic to the simultaneous centralizer*

$$\bigcap_{e \notin \text{forest}} C_G(c_e).$$

The assembled cover is regular if and only if every chord holonomy c_e belongs to the centre of G .

Proof. Set the source and target phase assignments equal in [theorem 4.2](#). Root elements defining deck maps are exactly those commuting with every c_e . Taking inverses accounts for the right-action composition convention and identifies the deck group with this centralizer. The assembled cover is connected and has degree $|G|$. The deck action on a fibre is free, so it is transitive exactly when the centralizer has order $|G|$, or equivalently is all of G . This is precisely the stated centrality condition. $\square$

Consequently every cyclic assembly in this model is globally regular. For D_m with $m \geq 3$, the permitted central holonomies are just the identity for odd m , and the identity and half-turn for even m . The degenerate groups D_1 and D_2 are abelian, so all their assemblies are globally regular. Marks can reduce the group of mark-preserving automorphisms without changing whether the underlying cover is regular; the criterion refers to the unmarked covering map.

Corollary 4.7 (A precise enumeration obstruction). *An algorithm that explicitly lists one record for each cyclic framed gluing class requires at least m^β records. In particular, $\beta = 1$ and $m = 2^b$ already give 2^b inequivalent classes, although each individual phase and each equivalence query use only polynomially many bits in b .*

This obstruction has a carefully limited meaning. It is not a lower bound for unknot recognition, nor even for deciding an arbitrary property of the assemblies. In fact, the connected underlying orientable surfaces have the same Euler characteristic and boundary counts for fixed block and port data; they need not be distinguished by unmarked surface homeomorphism. What grows is the number of *framed covering* classes. A hierarchy algorithm may be able to discard some of that data, but it must prove that doing so preserves the relevant attachments and continuations. For an explicit graph, $\beta \leq e$ and the binary orbit count has $O(e \log(m+1))$ bits, so exact exponentiation is polynomial in the listed input. If a standalone caller supplies an arbitrary binary integer β , the output itself can have exponentially many bits in $\log \beta$. The orbit-count API must then be interpreted as output-sensitive.

4.5 Surface topology without sheet expansion

For each connected component of the assembly graph, its assembled covering surface is connected because every local block is connected and every attachment is nonempty. If the base blocks are S_v , then

$$\chi(\tilde{S}) = |G| \sum_v \chi(S_v).$$

Gluing along complete boundary circles changes no Euler characteristic. An unused boundary port with peripheral element p contributes $|G|/\text{ord}(p)$ boundary components. Thus

$$b(\tilde{S}) = \sum_{\text{unused ports } j} \frac{|G|}{\text{ord}(p_j)}, \quad g(\tilde{S}) = \frac{2 - b(\tilde{S}) - \chi(\tilde{S})}{2}. \quad (17)$$

Right deck maps preserve the lifted orientation, and the base attachments reverse boundary orientations, so the assembled surface is orientable. All quantities in (17) are evaluated in binary. They help audit the input and output but do not replace the finer gluing equivalence conditions.

For a concrete nontrivial attachment, take two pair-of-pants bases with local monodromies T, R in D_m and glue their reflection ports. Each connected regular block cover has genus zero and $2m + 2$ boundary circles. The full preimages of the reflection ports consist of m circles, so the assembly has genus $m - 1$ and $2m + 4$ remaining boundary circles. All its phases are equivalent because its assembly graph is a tree. Thus even an assembly of exponentially large genus can have a small exact comparison problem when m is given in binary. The tests check this nontrivial peripheral gluing directly against expanded sheet permutations.

5 An independent compressed normal-disk certificate

5.1 The proposition certified by the interface

The input to this kernel is a finite face-paired tetrahedral triangulation $\mathcal{T}$ and a vector $x \in \mathbb{N}^{7t}$ of standard normal coordinates. The claimed conclusion is

$\mathcal{T}$ is a connected compact orientable 3-manifold with one torus boundary, and x represents a properly embedded disk whose boundary is essential in that torus.

This is a relative topological certificate. The input does not contain a knot diagram or a verified conversion from such a diagram. Even a genuine compressing disk does not identify an arbitrary torus-boundary manifold as a solid torus: $(S^1 \times D^2) \# (S^1 \times S^2)$ supplies a familiar counterexample. For an actual knot exterior in S^3 , a compressible boundary does imply the unknot; establishing that this is the supplied exterior is an additional obligation. The successful evidence object states its relative scope explicitly.

The producer accepts a *supplied* coordinate vector and tries to attach a small rank prime. It does not enumerate surfaces. A rejected vector means that this particular certificate format did not verify. It does not certify incompressibility or a nontrivial knot.

5.2 Finite triangulations are checked, not assumed

Tetrahedra and vertices are numbered. A face record is either a boundary marker or a destination tetrahedron together with a permutation of its four vertices; the image of the source face is determined by that permutation. The verifier reconstructs the quotient and checks the following conditions.

1. Every face pairing is reciprocal. A face is not glued to itself. All indices and permutations have the exact declared types and ranges.
2. The graph of tetrahedra joined through paired faces is connected. An orientation sign on every tetrahedron solves the face-orientation equations, whose multiplier is minus the sign of the gluing permutation.
3. Edge identifications never identify an edge with its own reverse. Each edge link has zero or two unpaired endpoints. Since its constituent segments are joined at endpoints, and an edge equivalence class gives a connected link, these links are circles or intervals.
4. Vertex links are reconstructed from their triangles, edges, and vertices. The preceding local checks make them connected surfaces; their Euler characteristics must be 2 when closed and 1 when they have boundary. They are consequently spheres or disks. Ideal vertices and nonmanifold vertices are rejected.
5. The finite boundary is connected and has Euler characteristic zero. The checked ambient orientation induces a boundary orientation, so this boundary is a torus.

The vertex-link conclusion uses the classification of compact surfaces. For a connected surface with nonempty boundary, Euler characteristic one forces a disk even without an orientability premise. The separate edge and link checks are essential: Euler characteristic alone does not certify that an arbitrary two-dimensional quotient is a surface.

These checks involve $O(t)$ combinatorial cells, represented by union-find structures and finite graphs. They are included in the measured certificate time. No field supplied by a topology engine is trusted as a substitute.

5.3 Normal coordinates and the support-rank certificate

The seven coordinates in a tetrahedron are ordered

$$(T_0, T_1, T_2, T_3, Q_{01|23}, Q_{02|13}, Q_{03|12}).$$

Triangles link vertices; the indicated quadrilateral avoids the two listed opposite edges. The verifier reconstructs the normal matching matrix A , checks $Ax = 0$, and checks the quadrilateral condition: at most one of the three quadrilateral coordinates is nonzero in each tetrahedron. Together with $x \geq 0$, these are the standard admissibility conditions for an embedded normal surface [3, 11].

Let $I = \text{supp } x$, $s = |I|$, and let A_I restrict the columns of A to I . Each face-arc matching row has two positive and two negative contributions, before cancellations caused by identifications. Triangle and quadrilateral columns are distinct, so its Euclidean norm is at most 2. This bound remains true after restricting columns.

Theorem 5.1 (Primitive support-nullity certificate). *Suppose $x \neq 0$ is admissible, $\gcd(x_1, \dots, x_{7t}) = 1$, and $\text{rank}_{\mathbb{Q}} A_I = s - 1$. Then the represented normal surface is connected.*

Proof. Write the coordinate vectors of its nonempty connected components as $x^{(1)}, \dots, x^{(k)}$. Each is a nonnegative integer vector satisfying the matching equations, supported in I , and their sum is x . The one-dimensional kernel of A_I therefore gives $x^{(j)} = \lambda_j x$, with $\lambda_j > 0$. Primitivity supplies integers b_i with $\sum_i b_i x_i = 1$, whence $\lambda_j = \sum_i b_i x_i^{(j)}$ is an integer. Since $\sum_j \lambda_j = 1$, there is exactly one component. $\square$

This is the classical extreme-ray argument, not a new general connectivity theorem for normal surfaces. The support condition is equivalent to the vector lying on an extreme ray of the nonnegative matching cone. It is stronger than merely being a connected normal disk. A connected disk outside this class can be rejected by the present format.

Lemma 5.2 (One prime certifies rational rank). *If a checked prime p satisfies $\text{rank}_{\mathbb{F}_p} A_I = s - 1$, then $\text{rank}_{\mathbb{Q}} A_I = s - 1$. Conversely, if the rational rank is $s - 1$, a suitable prime occurs among the first s primes.*

Proof. A nonzero minor modulo p is a nonzero integer minor. Thus the modular rank is a lower bound for rational rank. The exact nonzero vector x is in the rational kernel, giving the opposite upper bound $s - 1$.

For the converse, choose a nonzero $(s - 1)$ -minor D . Hadamard's inequality and the row-norm bound give $1 \leq |D| \leq 2^{s-1}$. A nonzero integer in this range has at most $s - 1$ distinct prime divisors: a product of s distinct primes is larger than 2^{s-1} . At least one of the first s primes does not divide D , and the modular rank at that prime is $s - 1$. When $s = 1$, the empty minor is 1 and rank zero is immediate. $\square$

The certificate stores the prime, not a trusted assertion that it is prime. The verifier restricts it to

$$2 \leq p \leq 32(s + 1)^2 \tag{18}$$

and proves primality by exact trial division. This deliberately generous bound keeps even that elementary test polynomial in the encoded input size. An adversary cannot turn verification into trial division of an arbitrary binary integer.

Here is an elementary supply proof for (18). Put $N = 32(s+1)^2$ and let $\pi(N)$ count primes at most N . The central binomial coefficient satisfies

$$\frac{2^N}{N+1} \leq \binom{N}{N/2} \leq N^{\pi(N)}.$$

For the upper bound, Legendre's formula shows that each term in the p -adic valuation is zero or one. Hence the full contribution of a prime p to the binomial coefficient is at most $p^{\lfloor \log_p N \rfloor} \leq N$. Consequently

$$\pi(N) \geq \frac{N - \log_2(N+1)}{\log_2 N} > s.$$

For completeness, $N - \log_2(N+1) \geq N/2$ here and $\log_2 N = 5 + 2\log_2(s+1) \leq 5 + 2s$; therefore the last quotient is at least $16(s+1)^2/(5+2s) > s$.

Corollary 5.3 (Coordinate size on the certified ray). *For a primitive vector with this support-nullity property, every positive coordinate is at most 2^{s-1} . The coordinates therefore have $O(t^2)$ total bits.*

Proof. Choose $s-1$ independent rows of A_I . Their signed maximal minors form a nonzero integer kernel vector. Dividing by their common divisor gives the primitive kernel generator, up to sign. Each minor has absolute value at most 2^{s-1} by the same row-norm estimate. There are at most $7t$ coordinates, each with $O(t)$ bits. $\square$

This is a certificate-size statement. It does not say that an algorithm can find the ray without examining exponentially many possible supports or quadrilateral choices.

5.4 Euler characteristic without expanding the surface

Let w_e denote the intersection number with a global triangulation edge. In local edge ab of tetrahedron i , it is

$$w_{ab} = T_a + T_b + Q_{01|23} + Q_{02|13} + Q_{03|12} - Q_{ab|cd},$$

where cd is the opposite edge. All local representatives of a global edge must give the same value; the verifier checks this directly. The normal arc count on face f is

$$a_f = \sum_{v \neq f} T_v + \sum_Q Q.$$

Counting each paired face once and each boundary face once gives

$$\begin{aligned} V_S &= \sum_{e \text{ global}} w_e, \\ E_S &= \sum_{f \text{ paired once}} a_f + \sum_{f \text{ boundary}} a_f, \\ F_S &= \sum_i x_i, \quad \chi(S) = V_S - E_S + F_S. \end{aligned}$$

These are the cells of the normal surface induced by the triangulation, computed through their binary multiplicities. The checker requires $\chi(S) = 1$ and a positive boundary-arc count. Combined with [theorem 5.1](#), the classification of surfaces gives a disk. No independent orientability test on the normal surface is required for this particular conclusion.

5.5 Essential boundary from intersection parity

The remaining disk could be boundary-parallel. To exclude it, assign to each boundary edge the parity

$$c(e) = w_e \bmod 2.$$

For every boundary triangle, the sum on its three edges is zero: each normal arc contributes two endpoints. Thus c is a cellular cocycle, the mod-2 Poincaré dual of the boundary multicurve. The implementation checks this identity as well as deriving the counts from the coordinates.

A cocycle on this finite cell structure is a coboundary precisely when there is a vertex potential ϕ satisfying

$$c(uv) = \phi(u) + \phi(v) \pmod{2}$$

on every boundary edge. A graph traversal assigns potentials and detects inconsistent equations. Loop edges are retained; on a one-vertex torus they can carry all the information.

Lemma 5.4 (Parity detects essential disk boundary on a torus). *Once the surface has been proved to be a disk, its boundary is essential in the checked torus if and only if c is not a coboundary.*

Proof. The disk has exactly one simple boundary curve. A null-homotopic simple curve in a torus has zero integral homology class. An essential simple curve has a primitive nonzero class $(a, b) \in \mathbb{Z}^2$. A primitive pair cannot have both entries even, so its image in $H_1(T^2; \mathbb{F}_2)$ is nonzero. Poincaré duality identifies that class with $[c]$ in cohomology. $\square$

The order of these conclusions matters. For an arbitrary disconnected normal surface, essential parallel boundary curves can cancel modulo two. Conversely, positive Euler characteristic alone can combine several components in misleading ways. Connectedness, disk type, and the single torus boundary are proved before parity is interpreted topologically. The code can calculate intermediate values in a different order, but it returns success only after every condition is satisfied.

Theorem 5.5 (Soundness and bit complexity of the disk verifier). *Acceptance of the face pairing, coordinate vector, and rank prime proves the relative proposition at the start of this section. With schoolbook integer arithmetic, a conservative bound for one verification is*

$$O(tB_x^2 + t^3 \log^2(t+2))$$

bit operations and polynomial space. No term is proportional to $\sum_i x_i$.

Proof. The finite combinatorial checks establish the ambient manifold and its boundary. Matching and quadrilateral admissibility establish the embedded normal surface. The modular rank and primitivity prove connectedness; Euler characteristic and boundary count prove disk type; the nonzero parity class proves essential boundary.

There are $O(t)$ coordinate and matching operations on B_x -bit integers. The gcd computations can be bounded by $O(tB_x^2)$. Dense elimination on an $O(t)$ by $O(t)$ matrix uses $O(t^3)$ operations modulo a prime of $O(\log(t+2))$ bits, including the lower-order pivot inversions and prime validation. The finite topology and parity traversals are smaller terms. All surface multiplicities are stored and manipulated as binary integers. $\square$

The implementation specializes $p = 2$ to packed exact XOR elimination. The producer may try s primes, so its rank work can be larger by that factor; the certificate verifier performs only one rank check. None of this replaces the more general orbit-counting algorithms of Agol–Hass–Thurston [4]. It is a compact certificate format for the primitive extreme case, with a small independent trusted computation.

5.6 An explicit exponential-weight fixture family

Let $F_0 = 0$, $F_1 = 1$, and $F_{r+2} = F_{r+1} + F_r$. Fibonacci growth in layered solid tori is classical [11]. We record a fully specified family because it makes the difference between coordinate size and expanded surface size directly reproducible.

For $t \geq 1$, number tetrahedra $0, \dots, t-1$ from the boundary inward. For $i < t-1$, face 0 of tetrahedron i is paired to tetrahedron $i+1$ by the permutation $(2, 1, 3, 0)$, and face 1 is paired by $(0, 3, 1, 2)$. A tuple here lists the images of vertices $0, 1, 2, 3$; it is not cycle notation. At the core, face 0 is glued to face 1 of the same tetrahedron by $(1, 2, 3, 0)$. Every inverse pairing is included. Faces 2 and 3 of tetrahedron 0 are the two boundary faces. These are the standard layered solid tori with meridian boundary intersection numbers $F_{t+1}, F_{t+2}, F_{t+3}$.

In tetrahedron i , put $r = t - i$ and set

$$x_i = (F_{r+1}, F_{r+1}, 0, 0, 0, F_r). \quad (19)$$

On this support, write the three nonzero variables as a_i, b_i, c_i . The core matching equations impose $a_{t-1} = b_{t-1} = c_{t-1}$. The interlayer equations impose

$$c_i = a_{i+1} = b_{i+1}, \quad a_i = a_{i+1} + c_{i+1}, \quad b_i = b_{i+1} + c_{i+1}.$$

They have a one-dimensional solution space over every field; all remaining restricted matching equations are satisfied by this recurrence. Hence the support has rank $3t - 1$ even modulo two. The core coordinate is one, so the vector is primitive. In particular the certificate prime can always be 2 for this family.

The exact number of elementary normal disks is

$$D_t = \sum_{r=1}^t (2F_{r+1} + F_r) = \sum_{r=1}^t F_{r+3} = F_{t+5} - 5. \quad (20)$$

The global triangulation edges have weights $F_2, F_3, \dots, F_{t+3}$: the core gives 1, 2, 3, and each new layer adds the next Fibonacci weight while retaining the previous global edge weights. Therefore

$$V_S = \sum_{k=2}^{t+3} F_k = F_{t+5} - 2.$$

The core paired face contributes 2 surface edges, the two paired faces at level $r \geq 2$ contribute $2F_{r+2}$, and the two outer boundary faces contribute $2F_{t+3}$. Hence

$$E_S = 2 + 2 \sum_{r=2}^t F_{r+2} + 2F_{t+3} = 2F_{t+5} - 8,$$

including the empty sum for $t = 1$. These formulas give $\chi(S) = 1$. There are $2F_{t+3} > 0$ boundary normal arcs. The boundary has one vertex and three edges. Its consecutive Fibonacci edge counts are not all even, so the parity cocycle cannot be a coboundary. This proves the certificate conditions for every t without an expansion.

Thus $D_t = \Theta(\varphi^t)$, where $\varphi = (1 + \sqrt{5})/2$, while the full coordinate vector has only $O(t^2)$ bits. The fixture generator is independent of Regina. The benchmark also compares the generated face pairings and coordinates with Regina's native layered-solid-torus construction for twelve small values, and retains the supplied inputs for all measured sizes.

At $t = 128$, the exact value of D_t is

$$2\,791\,715\,456\,571\,051\,233\,611\,642\,548.$$

The verifier manipulates the 896 coordinate positions, not that many individual disks. This is a test of certificate checking for a supplied triangulation, not a knot-diagram family with a demonstrated recognition speedup.

5.7 API and integration boundary

The module `fastunknot/normal_disk_certificate.py` exposes a producer, a Boolean verifier, and an evidence-producing audit function. A certificate contains exactly a version, the coordinate rows, and a rank prime. The audit reconstructs the matrix and every topological count; it does not accept a supplied rank, Euler characteristic, component count, or boundary-homology answer.

Cooperative cancellation exceptions propagate to the caller. Strict schema validation rejects Boolean values in integer positions and malformed face maps. Unit tests include valid disks, boundary-parallel vertex-linking disks, nonprimitive multiples, annuli, mismatched coordinates, conflicting quadrilaterals, invalid moduli, disconnected input, and relabelled copies.

The natural next integration is a transcript from a validated PD diagram to its finite compact exterior triangulation, together with a checker for that transcript. A producer may then be delegated to a fast topology engine while the positive answer is independently replayed. This work does not attach an unverified triangulation hash to a PD diagram and call that replay complete.

6 Reproducible experiments and negative evidence

6.1 Protocol and the meaning of a recorded time

The recorded environment is Python 3.12.14 on Linux x86-64, with Regina distribution 7.4.1 (engine version string 7.4) for the optional native comparisons. The measurements were made in a shared execution environment, not on a frequency-controlled dedicated benchmark machine. The artifacts retain individual measurements, random seeds, input data, result status, and source digests. Small timing differences should not be interpreted as stable effects.

The endpoint and normal-disk studies use five shuffled measured rounds after one excluded warmup. The gluing study uses seven shuffled rounds and an additional same-algorithm control. Every completed answer is checked outside the timing interval where the benchmark explicitly separates an oracle. For endpoint queries, the interval includes constructing a fresh word arena and performing the requested operation. Both arms use the same arena, relation-move code, and replayer; only the archived versus updated `CommonSubstring` implementation is exchanged.

Unless otherwise labelled, a time is the median of completed measurements for that arm. A paired ratio is the median of the within-round baseline/new ratios, so it need not equal the ratio of two separately computed medians. We give ratios only when every measured run of both arms completed. The raw endpoint JSON also records time spent reaching a resource limit; that field is elapsed time to interruption and is never treated here as a completion time.

All endpoint kernel and relation-operation cases use a shared work allowance of 2,000,000 and an arena-node allowance of 100,000. A resource failure is reported as `LIMIT`. It is neither an empty overlap set nor a negative knot decision. Work counts are implementation-defined cooperative charges, useful for explaining a particular change but distinct from the bit model in the proofs.

6.2 Endpoint progressions: completed cases and remaining failures

There are 27 kernel cases: nine families at binary exponent parameters $b = 8, 32, 500$. Except for the named fixed-marker controls, the repeated block and repetition count use $N = C = 2^b$. The archived baseline completes 13 of 27 cases under the fixed allowance; the rank-based version

completes 25 of 27. An independent containment, disjointness, and cardinality oracle checks each expected progression without enumerating its members.

Kernel, $b = 8$	Baseline (ms)	New (ms)	Paired ratio	Anchor q
65 marked blocks	148.225	0.574	290.4×	1
Single-letter marker power	231.578	0.711	318.8×	1
Binary bigram marker power	204.855	0.830	259.2×	2
Binary fourgram marker power	209.057	0.954	219.2×	4
Phase-compatible prefix	35.938	0.636	57.4×	1
Nonperiodic opposite word	232.038	0.799	290.4×	1
Rejected period, fallback	242.316	249.324	0.983×	—

Table 2: All five rounds complete in both arms for these rows. The last row records the cost of trying and rejecting the certificate, not an improvement. Times and ratios have different aggregation rules as stated in the protocol.

The new family is not limited to an expanded period of 32 or 64 individually checked occurrences. In the $b = 500$ power tests the period needs 502 bits and the occurrence count needs 501 bits. The following are completed new-arm measurements; the baseline reaches its work limit in every corresponding case, so no completion ratio is available.

Kernel, $b = 500$	New median (ms)	Charged work	Arena rules
Single-letter marker power	30.043	47,299	2,003
Binary bigram marker power	40.713	65,358	2,003
Binary fourgram marker power	51.875	97,977	2,006
Nonperiodic opposite word	38.093	61,831	2,507

Table 3: Binary-described progressions with exponentially many candidate lengths. Work and node counts are deterministic on these fixtures; timing is empirical. The three power rows each need one candidate-rank comparison; the nonperiodic-opposite row needs three.

The nonperiodic-opposite example is an informative ablation. An earlier, mathematically correct implementation evaluated the whole longest common extension in letter coordinates, then clipped the progression. At $b = 500$ it exhausted the same work allowance. The final implementation observes that only the largest candidate fails and settles the cutoff with three candidate equalities. Both the earlier implementation and its original benchmark harness are archived, with an explicit mapping from the historical source names to their archived copies. This separates the benefit of the progression theorem from the additional benefit of searching its ranks.

The damaged-period family deliberately has regularly spaced markers but unequal intervening blocks. The period certificate rejects it. At $b = 32$ and $b = 500$, both implementations still reach the work limit in the complete fallback. This is the two-case remainder in the 25/27 completion result. Regularly spaced anchors alone are not enough to accept an overlap formula.

The already successful short-period and 17-marker controls do not enter the new shortcut. Their work counts and arena-node counts match exactly between arms. The same holds for shared-subword LCS, shared-subword relator, and quotient controls. Their small wall-time variations provide context for the much larger activated effects; they are not reported as independent algorithmic improvements.

6.3 Downstream word operations improve, but not every large case

The 15 operation cases include three unaffected control families and two phase-shifted families at each exponent size. The baseline completes 10 cases; the new implementation completes 13. A cyclic-relator case includes rejection of a whole-donor move, finding a cyclic overlap, and applying the resulting move with exact checks. It is a local presentation operation, not an entire unknot-recognition run.

Operation	b	Baseline	New median	Endpoint hits
Phase-shifted LCS	8	613.197 ms	30.123 ms	2
Phase-shifted LCS	32	LIMIT	575.516 ms	2
Phase-shifted relator move	8	LIMIT	102.952 ms	8
Phase-shifted relator move	32	LIMIT	1,735.274 ms	8
Phase-shifted LCS	500	LIMIT	LIMIT	1
Phase-shifted relator move	500	LIMIT	LIMIT	1

Table 4: Exact operations downstream of the overlap query. Endpoint hits count successful local certificates even when subsequent work prevents the whole operation from completing. A LIMIT entry has no completion time.

For phase-shifted LCS at $b = 8$, the median paired ratio is 19.83. At $b = 32$, the relator operation completes with 1,866,801 charged work units; the archived letter-extension ablation still reaches the allowance. At $b = 500$, a successful endpoint query does not prevent later inherited matching and equality work from exhausting the budget. These retained failures motivate work on the exact comparison backend and amortized grammar growth, rather than a claim that all compressed relator operations are now inexpensive in practice.

6.4 Actual knot-diagram runs show no shortcut activation

The complete-recognition comparison uses twelve existing survivor diagrams, two mirrors, and the repository’s Gordian example. Inputs are rebuilt from their PD codes so that braid-source provenance cannot provide a hidden shortcut. All 15 cases return UNKNOT in both arms. Their certificate digests and recorded compressed-search statistic dictionaries agree, and the new endpoint trial and hit counts are zero throughout.

The sums of per-case median wall times are 1.649 seconds for the archived arm and 1.758 seconds for the updated arm. These sums are descriptive, not a single paired-trial statistic. The Gordian case contributes most of the time and has a broad timing spread despite identical executed search statistics. No end-to-end knot speedup is demonstrated, and this variation is not assigned to a shortcut that was never entered. The local positive results remain useful implementation advances, but a knot-level performance claim requires an activated knot-derived workload.

6.5 Regular-cover arithmetic at very large binary degrees

The independent literal-sheet oracle checks 3,480 comparisons: 3,410 equivalent and 70 inequivalent. These include 3,360 individual boundary-lift pair queries and 120 random marked graphs with loops, parallel edges, and disconnected components. Across them, 13,587 accepted literal gauge tuples agree with the compact output families. Exhaustive orbits on all 455 phase assignments of cycle rank two for C_m and D_m , $1 \leq m \leq 6$, agree with both the canonical keys and [theorem 4.5](#).

The timing family has 64 blocks and 32 named boundary lifts. Modulus bit length ranges from

39 to 24,007; a tree uses 63 edges and the cycle-constrained graph uses 80. Input generation and hashing are excluded. Full comparison includes raw-schema validation, monodromy and peripheral validation, marking validation, and solving. Prepared comparison starts from checked models.

24,007-bit modulus	Full comparison (ms)	Prepared (ms)	Family count
Tree with boundary constraints	5.564	1.279	1
Graph with cycle constraints	6.687	2.470	1

Table 5: Medians of seven rounds. The one tree family represents exactly 2^{23999} compatible isomorphisms, and the one cycle-constrained family represents two. No sheet expansion is attempted at these sizes.

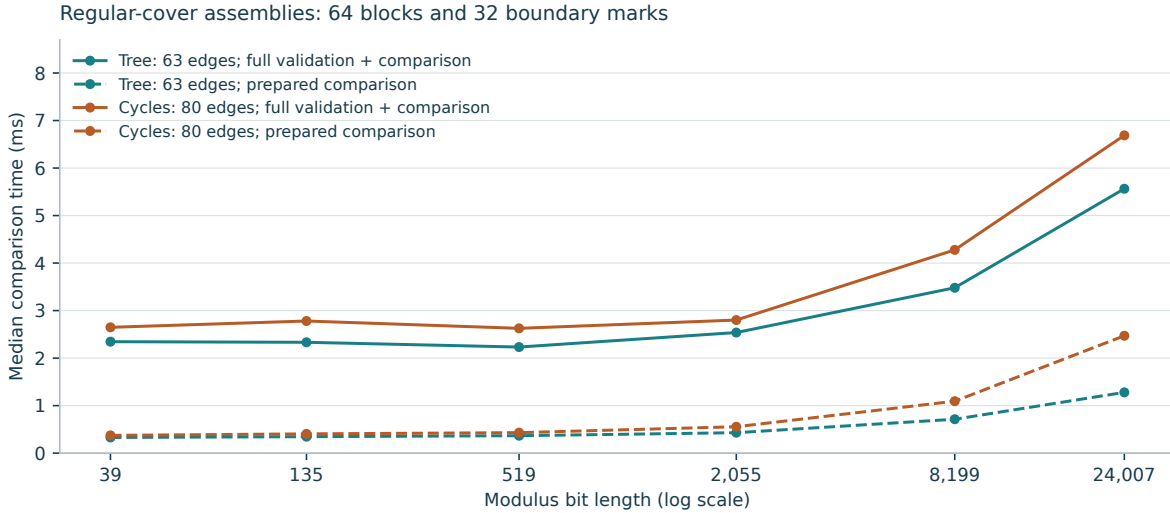


Figure 1: Observed gluing times as the binary modulus grows. Lines connect measured medians and are not fitted complexity laws. The same-algorithm control ratios range from 0.952 to 1.496; the study therefore supports feasibility on huge encoded degrees, not precise small-factor speed claims.

The exhaustive oracle is intentionally confined to small degree. There is no measured expanded baseline at a 24,007-bit modulus, and the article does not invent one by extrapolation. The proof supplies the reason the compact computation avoids sheet enumeration.

6.6 Normal disks: supplied witnesses of exponential weight

The independent checker accepts every saved Fibonacci fixture from one through 256 tetrahedra. For comparison, fresh worker processes call Regina’s `isCompressingDisc(True)` on the same supplied surfaces. The flag gives the native predicate the advantage of assuming connectedness; the independent checker proves it, and also validates the ambient triangulation. Native process startup, import, and input construction are excluded from its predicate clock. The native API documentation explicitly describes cutting and retriangulating, with large-coordinate memory costs [12]; this is a comparison against that API, not against the best known theoretical normal-surface algorithms.

Native workers have a three-second allowance and a 768 MiB address-space limit. The 22-tetrahedron failures are memory failures, not completed predicates at the limit. For 256 tetrahedra the maximum coordinate has 178 bits, and the complete serialized input plus

t	Elementary disks D_t	Checker (ms)	Native (ms)	Paired ratio
6	84	0.410	0.614	$1.36\times$
10	605	0.782	4.924	$7.06\times$
14	4,176	0.924	44.903	$49.37\times$
18	28,652	1.173	414.236	$353.25\times$
22	196,413	1.689	–	5/5 memory failures
32	24,157,812	2.090	–	not attempted
64	$\approx 1.177 \cdot 10^{14}$	4.168	–	not attempted
128	$\approx 2.792 \cdot 10^{27}$	9.778	–	not attempted
256	$\approx 1.571 \cdot 10^{48}$	24.943	–	not attempted

Table 6: Supplied normal-surface verification, not surface discovery or diagram-level recognition. Every checker row completes in all five rounds. The native 22-tetrahedron runs raise MemoryError under the 768 MiB process address-space cap. Exact large integers are retained in the fixtures.

certificate is 69,434 bytes. The interval includes reconstructing the finite manifold, matching, primitivity, rank, Euler characteristic, and boundary parity. No expansion proportional to D_t occurs.

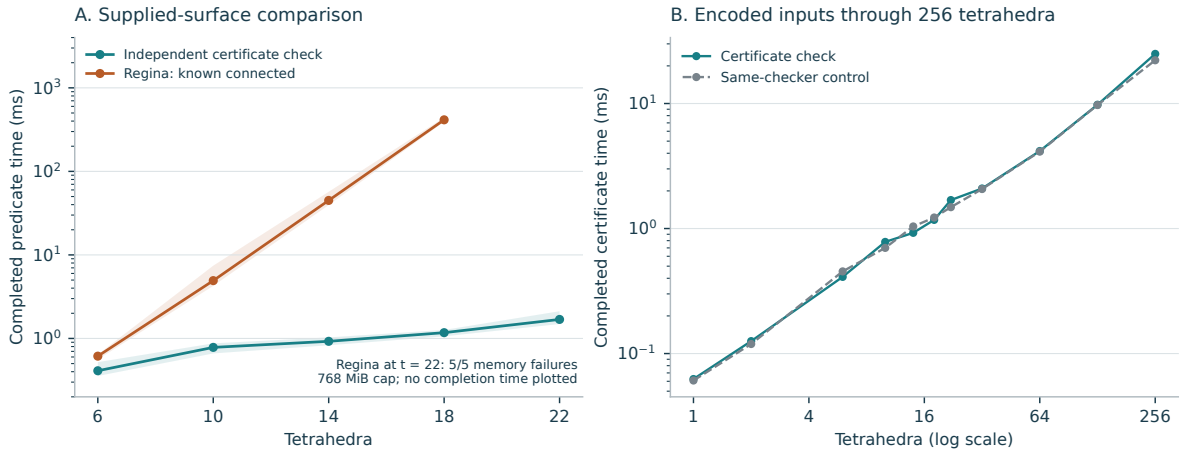


Figure 2: Normal-disk experiment. Panel A compares completed predicates; shading is the observed range of five samples, not a confidence interval. Panel B shows the checker and a same-checker control through the largest saved input. The proof, rather than a regression fitted to these points, establishes polynomial dependence on encoded input size.

The independent ambient-manifold audit samples 1,200 finite face-pairing inputs with one to five tetrahedra. The checker and Regina both accept 58 as connected compact orientable manifolds with one torus boundary, with zero mismatches. Accepted raw inputs are retained. Twelve native layered-solid-torus constructions agree with the independent fixture generator. In addition, 128 single-coordinate mutations are rejected, and boundary-parallel vertex-linking disks are rejected by the parity condition. These tests strengthen implementation confidence; the soundness theorem does not rest on the random sample.

6.7 Integrated validation and release structure

The final integrated test run passes all 735 tests in 110.513 seconds, with Regina enabled and no skipped tests. The complete log is included. The new endpoint, regular-cover, and normal-disk test modules contain 10, 14, and 12 tests, respectively; individual tests often perform exhaustive

or randomized comparisons over many inputs. Historical reference implementations needed by the maintained tests are included in their original relative paths.

Integration also exposed an existing worker-communication defect on the tested Python runtime. Retrying a timed `communicate(None)` could leave a large partially written stdin payload unsent, causing an otherwise simple worker to time out. The original 500,000-byte regression reproduced this failure. The correction makes one public `communicate(payload)` call in an I/O thread while the caller polls cancellation and the absolute deadline. Cleanup reaps the child, joins that thread, and closes the pipes. Two additional regressions cover large simultaneous output and cancellation with blocked input. This compatibility correction does not contribute to the measured kernel gains, and is identified separately in the patch.

The release contains the complete article source, its compiled PDF, figure sources and figures, the runnable maintained `fast/` code and tests, the necessary historical test oracles, new fixtures and raw measurements, an integration note, a patch against the pinned checkout, and a SHA-256 manifest. Generated interpreter caches and unrelated historical benchmark collections are excluded. The integration patch is checked against the pinned baseline before packaging.

The three measurement entry points are `fast/benchmark_compressed_endpoint.py`, `fast/regular_cover_research/reproduce.py`, and `fast/certificate_research/benchmark.py`. Figures can be rebuilt from the saved data without rerunning the benchmarks. The README records commands, optional dependencies, and the exact provenance of the baseline and archived ablation. This makes each claim independently inspectable and makes an unsuccessful reproduction report actionable.

7 What a quasi-polynomial recognition proof still requires

7.1 The specific missing theorem

The progress established here consists of exact local representations and operations. A proof of the requested global bound would need a theorem with the following shape: given a validated n -crossing knot diagram, a complete algorithm constructs and processes at most $2^{C(\log(n+2))^2}$ encoded objects, each of at most that many bits, and supplies independently checkable evidence for its terminal answer. The constants and the algorithm must be uniform over all diagrams. A reduction to another unbounded search does not satisfy the theorem.

For a hierarchy approach, the objects include the triangulations or handle structures, chosen normal surfaces, cut-open boundary patterns, gluing data, and any alternatives explored while finding or repairing a hierarchy. Controlling the number of successful cuts alone leaves the cost of each search and each representation conversion unresolved. For a compressed group approach, they include the presentation states, permitted relation moves, matching queries, and certificate transcripts. Polynomial exact word operations do not bound the length or breadth of that search.

The appropriate proof strategy must connect four statements: a complete choice principle, a bound on the choices actually explored, preservation of small encodings, and a checked terminal decision. The present work improves parts of the encoding and checking statements and isolates a quantitative obstacle for a natural choice space.

7.2 A measurable geometric entropy budget

Suppose an algorithm explicitly enumerates every joint framed phase-class assignment for independent cyclic assemblies simultaneously represented at one frontier, indexed by j . The

exact number of joint choices is

$$\prod_j m_j^{\beta_j} = 2^\Psi, \quad \Psi = \sum_j \beta_j \log_2 m_j.$$

For that particular enumeration strategy to fit the target, one needs $\Psi = O((\log n)^2)$, before charging the cost of each class. Merely proving that $\beta_j = O(\log n)$ is insufficient if a relevant $\log m_j$ can be $\Theta(n)$. Conversely, a degree m_j that is large as an integer is harmless for the number of phase classes on a tree, where $\beta_j = 0$ and all phases are removable by vertex gauges. Its bit length still contributes to the cost of representing and processing that class. The product counts simultaneous joint choices; the costs of separate tasks or successive stages instead add and must be accounted for separately.

This parameter makes an empirical investigation more discriminating than recording only genus, cover degree, or number of blocks. A hierarchy trace should record the actual frontier graph, the bits in its degrees, and its cycle rank after every simplification. It should also distinguish classes actually reached from all classes admitted by an unrestricted model. Our orbit formula does not establish that a knot hierarchy realizes the whole space.

Possible ways around the obstruction include proving that reachable frontiers satisfy a small entropy budget, or learning to operate on many phase classes symbolically, proving that all later tests preserve that symbolic description. Simple equality of canonical keys accomplishes neither by itself. A third possibility is a justified weaker equivalence, addressed next.

7.3 Equivalence must be justified by future operations

The framed covering equivalence used in [section 4](#) deliberately preserves the base projection. Forgetting that projection can identify different framed states: the underlying oriented surfaces may all have the same genus and boundary count. A smaller quotient could therefore be possible, but the quotient must retain the information used by the remaining algorithm.

A useful formulation is a continuation property. Two interface states may be merged when every allowed future attachment, surface test, and terminal recognition test has the same outcome after substituting either state. For an algorithm, one must give a finite computable sufficient criterion and prove that it is preserved by the permitted operations. Invoking homeomorphism of abstract surfaces without tracking the attaching curves, peripheral paths, and boundary pattern is not such a proof.

The new congruence solver gives a controlled laboratory for this question. All its currently supported restrictions can be kept as compact arithmetic families. One can ask exactly which added geometric operations force unions, nonlinear relations, or explicit splitting. That is a sharper next problem than assuming that a polynomial canonicalizer automatically yields a small state space.

8 Proposed research questions and concrete next milestones

The questions below are proposals, not consequences already established by the code. They are ordered by how directly they connect the present results to a demonstrable improvement of the complete recognizer. Each asks for a specific proof or experiment that can succeed or fail.

8.1 Priority A: connect exact local work to actual recognition

1. Can a PD-to-exterior conversion be independently replayed? Construct a finite compact triangulation from the validated diagram and export an elementary construction

transcript. The checker should recover the meridian and longitude data and prove the correspondence with the knot exterior. Alternatively, verify a sequence from a simple diagram-derived cell decomposition to the supplied triangulation. A triangulation digest proves only identity of bytes; it cannot replace this geometric argument. The immediate milestone is a positive knot certificate that combines this replay with [theorem 5.5](#) on a corpus of diagrams, including deliberately relabelled and corrupted transcripts. Size and running-time bounds must include the conversion transcript.

2. Which surviving knot computations activate the endpoint theorem? The present corpus does not establish a full recognition gain. Instrument compressed group search after every existing inexpensive filter, retaining the exact presentations that reach overlap matching. Record successful anchor size, progression step and count bits, comparison count, additional grammar nodes, and the reason the complete fallback was needed. The milestone is a paired recognition experiment on a predefined family with matching certificate hashes and nonzero shortcut activations. Constructed word examples remain useful tests, but should not be renamed as hard knots without a verified realization and an actual run through the recognizer.

3. Can a topology engine produce the new certificate economically? Given a disk found by the existing normal-surface backend, determine how often its standard vector satisfies the support-nullity condition. If it does not, investigate a checked reduction to a primitive extreme essential disk under precisely stated manifold assumptions. Record coordinate export, rank-prime discovery, conversion, and verification separately from surface search. Success would reduce reliance on a large producer for checking a positive result; it would not by itself accelerate discovery. A counterexample to an attempted extreme-disk reduction should become a permanent fixture.

4. Can modular rank checking reuse structured elimination? The current verifier is deliberately simple. A producer could furnish a sparse pivot transcript, a nonzero minor with elimination data, or a block decomposition derived from tetrahedron adjacency. The checker must verify the claimed rank, including all support columns, rather than merely replay unverified pivots. Compare time and certificate size against packed mod-2 elimination on layered examples, irregular triangulations, and examples whose first useful prime is odd. The milestone is a meaningful reduction on an independently chosen nonlayered family, with the same soundness statement and bounded verification work.

8.2 Priority B: remove the remaining compressed-word bottleneck

5. Can exact equality be accelerated with a rigorous bit bound? Rank galloping reduces the number of required equalities; the unresolved large phase-shifted operations still show that individual compressed comparisons matter. Evaluate recompression-based equality or pattern matching against the maintained implementation, using Jež's algorithm as a primary starting point [8]. Its word-RAM hypotheses must be translated to arbitrary binary positions and signed generator labels. The milestone is a total bound in reachable grammar size, height, and coordinate bit length, together with an implementation that preserves certificate replay and cancellation. Replacing exact comparison by an unchecked hash is not an acceptable shortcut.

6. Can occurrence progressions be generalized to a small union? An endpoint word may occur in a union of k arithmetic progressions even when no single progression suffices. Develop a grammar summary whose merges either produce a certified union of bounded size

or report that the representation has grown beyond a checked budget. For each component, state a period condition that makes candidate validity monotone, and prove that duplicate or overlapping candidates are handled exactly. The decisive issue is a bound on k along grammar construction and relation operations; allowing k to grow linearly with expanded length would lose the point. Start with two interleaved marker types and report adversarial families where the union count must increase.

7. Can anchor choice be adaptive without duplicating grammar work? The current finite list $q = 1, 2, 4, 8$ controls overhead. A structural selector might examine endpoint repetition, local grammar boundaries, or existing q -gram summaries to choose a larger useful anchor. Prove a bound for all failed anchor attempts as well as for the successful attempt. Shared summaries and reusable occurrence gaps are natural candidates. The milestone is an amortized bound in the final reachable grammar size and the total anchor material inspected, with negative benchmarks on words whose every short anchor has irregular occurrences.

8. Can galloping be extended across a sequence of relation moves? Successive cyclic reductions and overlaps often revisit related slices. Investigate certificates that survive a localized edit of the represented word. A maintained period certificate or occurrence summary must be invalidated exactly when its proof ceases to apply. Bound both the update cost and the growth of persistent grammar nodes over a complete transcript. The milestone is an amortized analysis for a specified move system and a verified implementation on saved knot-derived traces. A bound for one query cannot simply be multiplied by an assumed short move sequence.

8.3 Priority C: make geometric compression stable under hierarchy work

9. Can regular-cover blocks be extracted with certificates? The gluing kernel begins with connected regular blocks and explicit monodromy. Develop a recognizer for that structure inside the handle or boundary-pattern objects produced by the hierarchy. It must verify the local base, common fibre coordinates, surjectivity, based peripheral words, and every attachment phase. The milestone is an independently checked extraction from a real cut-open knot exterior. Failure to extract a block must preserve a complete fallback, not silently impose regularity.

10. Which cuts preserve the coherent attachment model? An entire boundary preimage glued by one right multiplication is much more structured than arbitrary independent permutations of its lifted circles. Classify the cuts, collapses, and hierarchy repairs that preserve this property. For operations that split a boundary preimage, ask whether the new pieces admit a small coset or interval description. Give either a closure theorem with an explicit size bound or a small obstruction showing why the current model is inadequate. Local regularity of the pieces must not be confused with global regularity of their assembly.

11. Can general transitive actions be handled without enumerating sheets? The regular model has a particularly simple centralizer: every equivariant map is a right multiplication. A transitive nonregular action has stabilizer data, and its equivariant automorphisms are controlled by a normalizer quotient. Seek a binary representation for those subgroups and attachment double-coset constraints in families actually arising from the hierarchy. The milestone is a proved polynomial algorithm for one strictly larger class, with a literal small-degree oracle. A formula for the group order alone does not supply an efficient algorithm for its action or transporter.

12. Can families of holonomies be propagated symbolically? For cyclic blocks, keep chord phases as affine subgroups or cosets in $(\mathbb{Z}/m)^\beta$ instead of listing m^β representatives. Test whether the geometric constraints needed by the hierarchy remain linear after forest changes, port gluing, and component elimination. For dihedral blocks, study whether parity variables can be retained as a compact algebraic system rather than branched independently. The milestone is closure under a specified list of operations, with a bound on the number and size of symbolic pieces. Counting or comparing one family is easier than proving that an arbitrary long computation stays within that family class.

13. Is there a useful weaker continuation equivalence? Determine which parts of the framed projection and peripheral data are actually observed by later hierarchy steps. Propose a computable quotient and prove the continuation property stated above. Then count its classes on the exact examples used for [theorem 4.5](#). A quotient that collapses all examples by forgetting information subsequently needed for a disk or annulus test does not qualify. Conversely, a sound quotient that removes the factor m^β would change the global search prospects much more than another constant-factor optimization of CRT.

14. Can the reachable entropy be bounded geometrically? Search for a relation between the frontier quantity Ψ , the original crossing number, and a monotone hierarchy complexity. A promising statement would prevent large degree-bit contributions from coexisting with many independent cycles, or would charge their creation to progress that cannot repeat. The milestone is a theorem for a nontrivial complete subclass of knots, followed by an explicit test of which hypothesis fails outside it. Bounds on genus, boundary count, or separator width should be translated into Ψ before claiming that they control phase enumeration.

8.4 Priority D: establish completeness and reproducible complexity evidence

15. How can essential surfaces be selected without exhaustive support search? The normal-disk verifier exposes how little work is needed once a suitable ray is supplied. Investigate whether geometric constraints permit a deterministic selection process that avoids the full quadrilateral-type and support search. Candidate approaches include a proved low-width decomposition of the relevant constraint system, a complete family of separators, or a structural reduction specific to knot exteriors. Each must bound failed choices and intermediate arithmetic, not just the size of one successful witness. The milestone is a complete theorem on a clearly defined class with unbounded crossing number and a counterexample search for each proposed extension.

16. Can compressed cutting be made representation-preserving end to end? Lackenby's framework includes efficient compressed operations for supplied surfaces [2]. The integration question is whether the particular encoding used by a complete implementation remains small after all boundary-pattern updates, simplifications, and subsequent surface requests. Prove a recurrence for actual encoded sizes, including names, paths, phase families, and certificates. The milestone is a checked multi-step hierarchy transcript whose size accounting survives every operation, together with a theorem explaining that accounting beyond the measured examples.

17. What is the right adversarial benchmark hierarchy? Maintain three separate levels: local binary-encoded stress families, verified geometric inputs, and complete knot-diagram runs. For every benchmark, record what is supplied and what is discovered. Include examples that defeat each structural shortcut, report UNKNOWN or resource limits explicitly, and retain the

artifacts that reproduce them. Paired runs, equivalence controls, and certificate hashes should be required before attributing a recognition gain to one kernel. The milestone is a benchmark release with a predefined scoring protocol and families that scale in crossing number, not just in an artificial repetition parameter.

18. Can the mathematical interface be formalized in small steps? The most suitable initial formalization targets are the endpoint clipping identity, monotone rank cutoff, forest gauge substitution, CRT family completeness, primitive-ray connectedness, and the boundary-parity lemma. Their statements have short, explicit premises and clear correspondences with certificate fields. A proof assistant development should distinguish the abstract theorem from the parser and integer implementation. The milestone is one end-to-end verified certificate checker, including input decoding and rejected malformed inputs, rather than only a formal version of an isolated algebraic identity.

8.5 Recommended order of work

The next implementation cycle should first finish PD-to-triangulation replay and identify actual knot traces that reach the improved overlap operation. Those tasks determine whether the new local capabilities improve the maintained recognizer on genuine inputs. In parallel, the highest-value theoretical work is the continuation quotient or symbolic propagation of holonomy families: the exact state-count result shows why comparison alone cannot settle that problem. Only after those interfaces are explicit should a global recurrence be proposed. It must include construction, all attempted choices, encoded state growth, and certificate replay. The present article therefore leaves the target bound open while supplying three concrete exact components, their proofs and failure boundaries, and experiments that another researcher can rerun or challenge.

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